PROJECTS · CAREERS · ROADMAP

Cyber Career Launchpad

The practical project and career guide for students and freshers: six entry-level roles and twelve projects to build.

06	12	0–1
Entry-level roles	**Hands-on projects**	**Years experience**
SOC, pentest, network, vulnerability, IR and GRC	Two per role, with milestones and validation	Built for students and freshers

JOIN THE COMMUNITY

Follow the WhatsApp channel

Updates, new project ideas and study resources. Scan the code or tap the button.

Open WhatsApp channel →

https://whatsapp.com/channel/0029VbDcJ0YF1YlUNvrCLp2d

PICK YOUR PATH

6 roles. 12 projects.

Choose one role, build its two projects in your own lab, and publish them on GitHub.

01 DETECT · TRIAGE · ESCALATE

SOC Analyst

Watch alerts. Spot real attacks. Escalate fast.

- Home SIEM Lab with Wazuh
- Phishing Email Analysis and Triage Playbook

02 ASSESS · VALIDATE · REPORT

Junior Penetration Tester / VAPT Analyst

Find weaknesses before attackers do.

- OWASP Juice Shop
- Network Pentest Lab

03 SEGMENT · FILTER · MONITOR

Network Security Engineer

Design and defend the network.

- Segmented Network with VLANs and Firewall ACLs
- Traffic Analysis and Intrusion Detection with Wireshark and Suricata

04 SCAN · PRIORITISE · VERIFY

Vulnerability Management Analyst

Scan, rank and fix vulnerabilities.

- Vulnerability Scan and Risk-Based Prioritisation Report
- Linux Hardening Baseline with CIS Benchmarks and Lynis

05 INVESTIGATE · PRESERVE · EXPLAIN

Incident Response / DFIR Analyst

Investigate incidents from the evidence.

- Windows Endpoint Detection Lab with Sysmon
- Digital Forensics Case

06 ASSESS · DOCUMENT · IMPROVE

GRC / Security Compliance Analyst

Turn risk into clear decisions.

- Security Risk Assessment and Risk Register for a Fictional Company
- Policy Pack and Control Mapping

> **Safety first:** use isolated labs and test only systems you own or are authorised to test.

ROLE 01 · DETECT · TRIAGE · ESCALATE

SOC Analyst (Tier 1)

01

Watch alerts. Spot real attacks. Escalate fast.

Log analysis | SIEM triage | TCP/IP & DNS basics | MITRE ATT&CK | Ticketing & shift handover

PROJECT 1

Home SIEM Lab with Wazuh

Beginner–Intermediate | ⏱ 15–20 h | Wazuh (all-in-one) | Ubuntu Server VM | Windows eval VM (optional)

Build a small, isolated monitoring lab, collect logs from endpoints, write a detection for failed-login bursts and triage the resulting alerts like a Tier 1 analyst.

Test source (Your own VM / script) ▶ **Target endpoint** (Ubuntu + Wazuh agent) ▶ **Wazuh server** (Manager · indexer) ▶ **Dashboard** (Alerts & triage)

Build it in 5 steps

1. Install Wazuh all-in-one following the official quickstart
2. Generate benign test events in your own lab
3. Tune or write one custom rule and map alerts to MITRE ATT&CK T1110
4. Build a dashboard and triage three alerts with true/false-positive reasoning
5. Write the README and an incident-style triage report

> **You'll ship:** README with architecture diagram and setup notes · Sanitised custom rule file and config notes · Dashboard screenshots (no secrets or real IPs)

PROJECT 2

Phishing Email Analysis and Triage Playbook

Beginner | ⏱ 10–14 h | Text editor / Thunderbird (.eml) | Google Admin Toolbox: Messageheader | MXToolbox

Analyse email headers, authentication results and links from sample messages you create or obtain from public training sets, then turn your method into a repeatable playbook.

Suspicious .eml (Sample email) ▶ **Header review** (SPF · DKIM · DMARC) ▶ **URL / file check** (Reputation & sandbox) ▶ **IOC list** (Defanged indicators) ▶ **Verdict** (Actions & comms)

Build it in 5 steps

1. Learn header fields
2. Check URLs and file hashes via reputation services without opening them
3. Extract and defang indicators
4. Draw the playbook flowchart, decision criteria and user-reply template
5. Write up each sample with verdict and recommended action

> **You'll ship:** Playbook (Markdown + flowchart PNG) · Six analysis write-ups · IOC table in CSV (defanged)

ROLE 02 · ASSESS · VALIDATE · REPORT

Junior Penetration Tester / VAPT Analyst

02

Find weaknesses before attackers do.

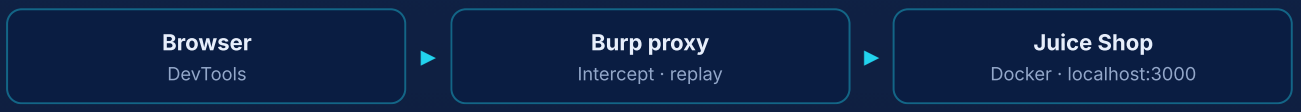
HTTP & web basics | OWASP Top 10 | Nmap & enumeration | Burp Suite | CVSS scoring

PROJECT 1

OWASP Juice Shop: Web Application Security Assessment

Intermediate | ⏱ 20–25 h | Docker | OWASP Juice Shop | Burp Suite Community

Treat a locally hosted, intentionally vulnerable web app as a client engagement: define scope, test methodically, score findings and deliver a professional report.



Build it in 5 steps

1. Write the scope and rules-of-engagement document
2. Test at least six issues across OWASP Top 10 categories
3. Document each finding
4. Write remediation advice in developer-friendly language
5. Assemble the final report and repository

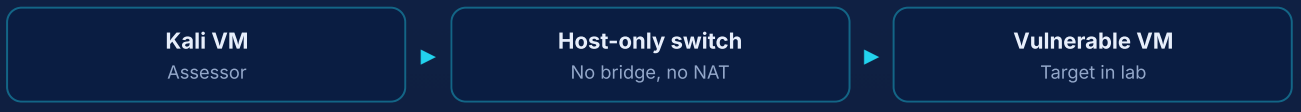
You'll ship: Report PDF: executive summary, scope, method, findings, appendix · Rules-of-engagement document · Evidence screenshots with secrets removed

PROJECT 2

Network Pentest Lab: Enumerate and Report on a Vulnerable Host

Intermediate | ⏱ 20–25 h | Kali Linux | Metasploitable 2 (or similar intentionally vulnerable VM) | Nmap

Run a structured assessment against a deliberately vulnerable virtual machine in a closed network, validate a few findings safely and write a prioritised report.



Build it in 5 steps

1. Write scope and rules of engagement
2. Enumerate four or five services and research versions in NVD
3. Validate three or four findings with minimal-impact proof
4. Write the report with risk ratings and remediation
5. Revert snapshots and document clean-up

You'll ship: Sanitised scan outputs · Findings table with severity and evidence · Report PDF

ROLE 03 · SEGMENT · FILTER · MONITOR

Network Security Engineer (Junior)

03

Design and defend the network.

Subnetting & routing | VLANs & ACLs | Firewall rules | Wireshark | IDS/IPS basics

PROJECT 1

Segmented Network with VLANs and Firewall ACLs

Beginner–Intermediate | ⏱ 15–20 h | Cisco Packet Tracer or GNS3 | pfSense CE (optional) | diagrams.net

Design a small-company network with staff, server and guest zones, enforce a default-deny policy and prove every rule with a test matrix.



Build it in 5 steps

1. Build the topology with VLANs and trunks
2. Write ACLs
3. Add DHCP, NAT and port security on access ports
4. Run the test matrix
5. Export sanitised configs and document

You'll ship: Topology file (.pkt / GNS3) and diagram · Sanitised configuration files · Test matrix table

PROJECT 2

Traffic Analysis and Intrusion Detection with Wireshark and Suricata

Intermediate | ⏱ 15–20 h | Wireshark | Suricata | Ubuntu VM

Learn to read packets, then replay public sample captures through Suricata and write and tune your own detection rules.



Build it in 5 steps

1. Analyse three sample PCAPs covering DNS, HTTP and a TLS handshake
2. Enable Emerging Threats Open rules with suricata-update
3. Write three custom rules
4. Validate and tune thresholds against clean and noisy captures
5. Write rule rationale and README

You'll ship: Annotated packet analysis notes · Custom .rules file with comments · Sanitised eve.json excerpts

ROLE 04 · SCAN · PRIORITISE · VERIFY

Vulnerability Management Analyst

04

Scan, rank and fix vulnerabilities.

Scanner operation | CVSS & CVE/NVD | Risk-based prioritisation | Linux/Windows hardening | Spreadsheet reporting

PROJECT 1

Vulnerability Scan and Risk-Based Prioritisation Report

Beginner–Intermediate | ⏱ 15–20 h | Greenbone Community Edition | Ubuntu / Windows lab VMs | NVD & CVSS calculator

Scan a small lab, compare authenticated and unauthenticated results, enrich findings with CVSS and KEV data and fix the highest-risk issues.

Asset inventory (Owner · criticality) ▶ **Greenbone scan** (Authenticated) ▶ **Enrichment** (NVD · CVSS · KEV) ▶ **Prioritise** (Ticket · owner) ▶ **Rescan** (Verify closed)

Build it in 5 steps

1. Deploy Greenbone following the official documentation
2. Triage
3. Document a simple prioritisation model
4. Remediate top issues on the lab VMs
5. Rescan and report before/after metrics

You'll ship: Asset register · Sanitised scan reports · Prioritised remediation tracker

PROJECT 2

Linux Hardening Baseline with CIS Benchmarks and Lynis

Beginner | ⏱ 10–15 h | Ubuntu Server VM | Lynis | CIS Benchmark (free download)

Create a repeatable hardening script for an Ubuntu server, justify each control and measure the effect with before-and-after audits.

Default Ubuntu (Fresh snapshot) ▶ **Baseline audit** (Lynis report) ▶ **Hardening script** (Idempotent Bash) ▶ **Re-audit** (Lynis again) ▶ **Evidence** (Mapping table)

Build it in 5 steps

1. Study CIS Level 1 sections
2. Implement them in an idempotent script
3. Re-run Lynis and verify required services still work
4. Add rollback notes and exception handling
5. Write README and mapping table

You'll ship: Hardening script · Mapping: CIS ID → change → rationale → verification command · Before/after audit summary

ROLE 05 · INVESTIGATE · PRESERVE · EXPLAIN

Incident Response / DFIR Analyst (Junior)

05

Investigate incidents from the evidence.

Windows & Linux internals · Event log analysis · Evidence handling · Timeline building · Memory & disk forensics basics

PROJECT 1

Windows Endpoint Detection Lab with Sysmon

Intermediate · ⏱ 15–20 h · Windows 10/11 evaluation VM · Sysmon + community config · PowerShell / Event Viewer

Generate safe, well-known attacker-style behaviour in an isolated Windows VM, find the evidence in Sysmon logs and write detection logic around it.

Atomic test (Isolated VM) ▶ **Sysmon + logs** (Event IDs 1 · 3 · 11 · 13) ▶ **Analysis** (PowerShell queries) ▶ **Mapping** (ATT&CK Navigator) ▶ **Detection** (Rule + timeline)

Build it in 5 steps

1. Install Sysmon with a community configuration
2. Locate each artifact in Sysmon and Windows events
3. Write detection logic
4. Run clean-up and reset the VM
5. Write the incident timeline and README

You'll ship: ATT&CK coverage table / Navigator layer · Detection rules with comments · Annotated event screenshots

PROJECT 2

Digital Forensics Case: Disk and Memory Analysis

Intermediate · ⏱ 20–25 h · Autopsy / Sleuth Kit · Volatility 3 · SANS SIFT Workstation (free VM)

Work a public forensic scenario end to end: preserve evidence, analyse disk and memory images, build a timeline and write a factual report.

Evidence image (Public dataset) ▶ **Hash & verify** (Read-only copy) ▶ **Analysis** (Autopsy · Volatility) ▶ **Timeline** (Correlated events) ▶ **Report** (Facts vs inference)

Build it in 5 steps

1. Set up SIFT/Autopsy
2. File-system analysis
3. Memory analysis
4. Build a combined timeline
5. Write the report separating facts from inferences

You'll ship: Case report · Timeline CSV · Chain-of-custody form

ROLE 06 · ASSESS · DOCUMENT · IMPROVE

GRC / Security Compliance Analyst (Junior)

06

Turn risk into clear decisions.

Risk assessment · Control frameworks · Policy writing · Evidence collection · Spreadsheet modelling

PROJECT 1

Security Risk Assessment and Risk Register for a Fictional Company

Beginner · ⏱ 12–18 h · Excel or Google Sheets · NIST CSF 2.0 · NIST SP 800-30

Invent a realistic small company, identify its assets and threats, score risks consistently and produce a treatment plan an executive could act on.

Scope & assets (Systems · data) ▶ **Threats** (Scenarios) ▶ **Scoring** (Likelihood × impact) ▶ **Risk register** (Owners · status) ▶ **Treatment** (Mitigate · accept)

Build it in 5 steps

1. Build the asset inventory and data classification
2. Define a 5×5 scoring method
3. Score risks and build a heat map
4. Write the treatment plan with owners and dates
5. Create a one-page executive summary and README

You'll ship: Risk register (xlsx/csv) · Heat map · Treatment plan

PROJECT 2

Policy Pack and Control Mapping

Beginner–Intermediate · ⏱ 15–20 h · CIS Controls v8 (free) · NIST CSF 2.0 · Spreadsheet

Write four short security policies, map IG1 safeguards to NIST CSF 2.0 functions and produce a gap assessment with a 90-day roadmap.

Policies (AUP · access · IR · backup) ▶ **Control mapping** (CIS ↔ CSF) ▶ **Gap assessment** (Implemented · partial · none) ▶ **Roadmap** (90-day plan)

Build it in 5 steps

1. Review the 18 CIS Controls and the IG1 safeguards
2. Self-assess the fictional company
3. Prioritise the gaps with a stated rationale
4. Build the 90-day roadmap
5. Write README

You'll ship: Policy documents (md/pdf) · Mapping workbook · Gap report

DON'T STOP HERE

Build. Document. Get hired.

JOIN THE COMMUNITY

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New project ideas, updates and study resources.

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